

Elasticsearch - Text Mapping and Analysis

Ensure kibana is up

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ docker-compose ps
      Name                    Command                                State      Ports
-----
es-container /bin/tini -- /usr/local/bi ...      Exit 143
kb-container /bin/tini -- /usr/local/bi ...      Exit 0

ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ docker-compose up -d kibana
Building with native build. Learn about native build in Compose here: https://docs.docker.com/go/compose-native-build/
Starting es-container ... done
Starting kb-container ... done

ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ docker-compose logs -f kibana
Attaching to kb-container
kb-container | [2025-07-21T14:33:24.178+00:00][INFO ][plugins-service] Plugin "metricsEntities" is disabled.
kb-container | [2025-07-21T14:33:24.773+00:00][INFO ][http.server.Preboot] http server running at http://0.0.0.0:5601
kb-container | [2025-07-21T14:33:26.057+00:00][INFO ][plugins-system.preboot] Setting up [1] plugins: [interactiveSetup]
kb-container | [2025-07-21T14:33:26.076+00:00][INFO ][preboot] "interactiveSetup" plugin is holding setup: Validating Ela
sticsearch connection configuration...
kb-container | [2025-07-21T14:33:26.265+00:00][INFO ][root] Holding setup until preboot stage is completed.
kb-container |
kb-container | i Kibana has not been configured.
kb-container |
kb-container | Go to http://0.0.0.0:5601/?code=428760 to get started.
kb-container |
kb-container | [2025-07-21T14:34:17.473+00:00][WARN ][config.deprecation] The default mechanism for Reporting privileges
will work differently in future versions, which will affect the behavior of this cluster. Set "xpack.reporting.roles.enabled"
to "false" to adopt the future behavior before upgrading.
kb-container | [2025-07-21T14:34:18.475+00:00][INFO ][plugins-system.standard] Setting up [112] plugins: [translations,li
censing,globalSearch,globalSearchProviders,features,mapsEms,licenseApiGuard,usageCollection,taskManager,telemetryCollectionMa
nager,telemetryCollectionXpack,kibanaUsageCollection,sharedUX,share,embeddable,uiActionsEnhanced,screenshotMode,screenshottin
g,banners,telemetry,newsfeed,fieldFormats,expressions,dataViews,charts,esUiShared,bfetch,data,savedObjects,presentationUtil,e
xpressionShape,expressionRevealImage,expressionRepeatImage,expressionMetric,expressionImage,customIntegrations,home,searchpro
filer,painlessLab,grokdebugger,management,watcher,licenseManagement,advancedSettings,spaces,security,savedObjectsTagging,repo
rting,lists,fileUpload,ingestPipelines,encryptedSavedObjects,dataEnhanced,cloud,snapshotRestore,eventLog,actions,alerting,tri
ggersActionsUi,transform,stackAlerts,ruleRegistry,savedObjectsManagement,console,controls,graph,fleet,indexManagement,remoteC
lusters,crossClusterReplication,indexLifecycleManagement,visualizations,canvas,visTypeXy,visTypeVislib,visTypeVega,visTypeTim
eseries,rollup,visTypeTimeline,visTypeTagcloud,visTypeTable,visTypeMetric,visTypeHeatmap,visTypeMarkdown,dashboard,maps,dashb
oardEnhanced,expressionTagcloud,expressionPie,visTypePie,expressionMetricVis,expressionHeatmap,expressionGauge,dataViewFieldE
ditor,lens,cases,timelines,discover,osquery,observability,discoverEnhanced,dataVisualizer,ml,uptime,securitySolution,infra,up
gradeAssistant,monitoring,logstash,enterpriseSearch,apm,dataViewManagement]
```

Open in browser and navigate to Dev Tools

The screenshot shows the Elastic Kibana web interface. On the left, a large 'Welcome to Elastic' banner features a colorful logo and a call to action: 'Start by adding integrations'. Below this, there's a link to 'Explore on my own'. The right sidebar contains a navigation menu with categories: 'APM', 'Uptime', 'User Experience', 'Security' (expanded), 'Management' (expanded), and 'Integrations'. Under 'Security', 'Dev Tools' is highlighted. Under 'Management', 'Dev Tools' is also visible. The top of the sidebar shows the 'elastic' logo and a 'Home' button.

Mapping:

Inspect existing (implicit) mapping

The screenshot shows the Elasticsearch console interface. The top navigation bar includes 'Console', 'Search Profiler', 'Grok Debugger', 'Painless Lab', and 'BETA'. A 'Click to send request' button is visible. The 'History' tab is active, showing a list of requests. The first request is a GET to /products/_mapping. The response is a JSON object showing the mapping for the 'products' index. The response is highlighted in green, indicating a successful status (200 - OK) and a response time of 201 ms.

```
1 GET /products/_mapping
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
```

```
{
  "products": {
    "mappings": {
      "properties": {
        "in_stock": {
          "type": "long"
        },
        "name": {
          "type": "text",
          "fields": {
            "keyword": {
              "type": "keyword",
              "ignore_above": 256
            }
          }
        },
        "price": {
          "type": "long"
        },
        "stock": {
          "type": "long"
        },
        "tags": {
          "type": "text",
          "fields": {
            "keyword": {
              "type": "keyword",
              "ignore_above": 256
            }
          }
        }
      }
    }
  }
}
```

Explicit mapping

The screenshot shows the Elasticsearch console interface. The top navigation bar includes 'Console', 'Search Profiler', 'Grok Debugger', 'Painless Lab', and 'BETA'. A 'Click to send request' button is visible. The 'History' tab is active, showing a list of requests. The first request is a PUT to /masterclass. The response is a JSON object showing the mapping for the 'masterclass' index. The response is highlighted in green, indicating a successful status (200 - OK) and a response time of 710 ms.

```
1 GET /products/_mapping
2
3 PUT /masterclass
4
5 {
6   "mappings": {
7     "properties": {
8       "theme": {
9         "type": "text"
10      },
11      "duration": {
12        "type": "integer"
13      },
14      "level": {
15        "type": "text"
16      },
17      "animator.first_name": {
18        "type": "text"
19      },
20      "animator.last_name": {
21        "type": "text"
22      },
23      "animator_email": {
24        "type": "keyword"
25      }
26    }
27  }
28 }
```

```
{
  "acknowledged": true,
  "shards_acknowledged": true,
  "index": "masterclass"
}
```

Ingest the CSV with Python, bulk helper

Download data

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ wget https://dst-de.s3.eu-west-3.amazonaws.com/elasticsearch_fr/datasets/EmployerReviews.csv
--2025-07-23 11:38:34-- https://dst-de.s3.eu-west-3.amazonaws.com/elasticsearch_fr/datasets/EmployerReviews.csv
Resolving dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)... 3.5.204.164, 3.5.205.184
Connecting to dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)|3.5.204.164|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 38738315 (37M) [text/csv]
Saving to: 'EmployerReviews.csv'

EmployerReviews.csv 100%[=====] 36.94M 36.6MB/s in 1.0s

2025-07-23 11:38:35 (36.6 MB/s) - 'EmployerReviews.csv' saved [38738315/38738315]
```

Create bulk_script.py

Give rights to the bulk_script.py file and install Python library

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ sudo chmod +x ./bulk_script.py
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ pip3 install elasticsearch
Collecting elasticsearch
  Downloading elasticsearch-9.0.2-py3-none-any.whl (914 kB)
    | 914 kB 8.6 MB/s
Collecting typing-extensions
  Downloading typing_extensions-4.13.2-py3-none-any.whl (45 kB)
    | 45 kB 2.6 MB/s
Collecting python-dateutil
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
    | 229 kB 13.6 MB/s
Collecting elastic-transport<9,>=8.15.1
  Downloading elastic_transport-8.17.1-py3-none-any.whl (64 kB)
    | 64 kB 2.8 MB/s
Requirement already satisfied: six>=1.5 in /usr/lib/python3/dist-packages (from python-dateutil->elasticsearch) (1.14.0)
Collecting urllib3<3,>=1.26.2
  Downloading urllib3-2.2.3-py3-none-any.whl (126 kB)
    | 126 kB 16.7 MB/s
Requirement already satisfied: certifi in /usr/lib/python3/dist-packages (from elastic-transport<9,>=8.15.1->elasticsearch) (2019.11.28)
Installing collected packages: typing-extensions, python-dateutil, urllib3, elastic-transport, elasticsearch
Successfully installed elastic-transport-8.17.1 elasticsearch-9.0.2 python-dateutil-2.9.0.post0 typing-extensions-4.13.2 urllib3-2.2.3
```

Remove wrong client and install correct one

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ pip3 uninstall -y elasticsearch elastic-transport
Found existing installation: elasticsearch 9.0.2
Uninstalling elasticsearch-9.0.2:
  Successfully uninstalled elasticsearch-9.0.2
Found existing installation: elastic-transport 8.17.1
Uninstalling elastic-transport-8.17.1:
  Successfully uninstalled elastic-transport-8.17.1
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ pip3 install "elasticsearch>=8,<9"
Collecting elasticsearch<9,>=8
  Downloading elasticsearch-8.18.1-py3-none-any.whl (906 kB)
    | 906 kB 5.9 MB/s
Requirement already satisfied: python-dateutil in /home/ubuntu/.local/lib/python3.8/site-packages (from elasticsearch<9,>=8) (2.9.0.post0)
Collecting elastic-transport<9,>=8.15.1
  Using cached elastic_transport-8.17.1-py3-none-any.whl (64 kB)
Requirement already satisfied: typing-extensions in /home/ubuntu/.local/lib/python3.8/site-packages (from elasticsearch<9,>=8) (4.13.2)
Requirement already satisfied: six>=1.5 in /usr/lib/python3/dist-packages (from python-dateutil->elasticsearch<9,>=8) (1.14.0)
Requirement already satisfied: certifi in /usr/lib/python3/dist-packages (from elastic-transport<9,>=8.15.1->elasticsearch<9,>=8) (2019.11.28)
Requirement already satisfied: urllib3<3,>=1.26.2 in /home/ubuntu/.local/lib/python3.8/site-packages (from elastic-transport<9,>=8.15.1->elasticsearch<9,>=8) (2.2.3)
Installing collected packages: elastic-transport, elasticsearch
Successfully installed elastic-transport-8.17.1 elasticsearch-8.18.1
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ pip3 install "urllib3<2"
Collecting urllib3<2
  Downloading urllib3-1.26.20-py2.py3-none-any.whl (144 kB)
    | 144 kB 5.4 MB/s
Installing collected packages: urllib3
  Attempting uninstall: urllib3
    Found existing installation: urllib3 2.2.3
    Uninstalling urllib3-2.2.3:
      Successfully uninstalled urllib3-2.2.3
Successfully installed urllib3-1.26.20
```

Run the bulk_script.py file

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/elasticsearch$ python3 bulk_script.py
/usr/lib/python3/dist-packages/requests/__init__.py:89: RequestsDependencyWarning: urllib3 (1.26.20) or chardet (3.0.4) doesn't match a supported version!
  warnings.warn("urllib3 ({}) or chardet ({}) doesn't match a supported version"
bulk_script.py:11: DeprecationWarning: Passing transport options in the API method is deprecated. Use 'Elasticsearch.options()' instead.
  helpers.bulk(es, reader, index = "reviews", chunk_size = 5000, request_timeout = 120)
```

Return to Elasticsearch UI to display mapping of new imported index

The screenshot shows the Elasticsearch UI Console with the 'Console' tab selected. The left pane displays the mapping for the 'reviews' index, which was just imported. The right pane shows the resulting JSON mapping structure. A tooltip 'Click to send request' is visible over the 'GET /reviews/_mapping' request.

```
1 GET /products/_mapping
2
3 PUT /masterclass
4 {
5   "mappings": {
6     "properties": {
7       "theme": {
8         "type": "text"
9       },
10      "duration": {
11        "type": "integer"
12      },
13      "level": {
14        "type": "text"
15      },
16      "animator.first_name": {
17        "type": "text"
18      },
19      "animator.last_name": {
20        "type": "text"
21      },
22      "animator.email": {
23        "type": "keyword"
24      }
25    }
26  }
27 }
28
29 GET /reviews/_mapping
30
```

```
6     "type": "text",
7     "fields": {
8       "keyword": {
9         "type": "keyword",
10        "ignore_above": 256
11      }
12    },
13  },
14  "Date": {
15    "type": "text",
16    "fields": {
17      "keyword": {
18        "type": "keyword",
19        "ignore_above": 256
20      }
21    }
22  },
23  "Rating": {
24    "type": "text",
25    "fields": {
26      "keyword": {
27        "type": "keyword",
28        "ignore_above": 256
29      }
30    }
31  },
32  "ReviewTitle": {
33    "type": "text",
34    "fields": {
35      "keyword": {
36        "type": "keyword",
37        "ignore_above": 256
38      }
39    }
40  }
41 }
42 }
43 }
44 }
```

Display some documents from the index

The screenshot shows the Elasticsearch UI Console with the 'Console' tab selected. The left pane displays the mapping for the 'reviews' index. The right pane shows the resulting JSON mapping structure. A tooltip 'Click to send request' is visible over the 'GET /reviews/_mapping' request. The bottom pane shows the search results for the 'reviews' index.

```
1 GET /products/_mapping
2
3 PUT /masterclass
4 {
5   "mappings": {
6     "properties": {
7       "theme": {
8         "type": "text"
9       },
10      "duration": {
11        "type": "integer"
12      },
13      "level": {
14        "type": "text"
15      },
16      "animator.first_name": {
17        "type": "text"
18      },
19      "animator.last_name": {
20        "type": "text"
21      },
22      "animator.email": {
23        "type": "keyword"
24      }
25    }
26  }
27 }
28
29 GET /reviews/_mapping
30
```

```
101 technical concept of my field ,which is very necessary
102 for a fresher and of course a first to move strongly
103 towards a bright future.",
104 "Rating" : "5",
105 "Date" : "24-01-2018"
106 }
107 {
108   "_index" : "reviews",
109   "_id" : "YHYfNSgBQuaETKQjIkME",
110   "_score" : 1.0,
111   "_source" : {
112     "ReviewTitle" : "Free and fun",
113     "CompleteReview" : "Work freely without any interference
114     of higher authority , learn lot about that feild have
115     good guidance by senior people, but there is no growth
116     in that company",
117     "Rating" : "3",
118     "Date" : "22-01-2018"
119   }
120 }
121 {
122   "_index" : "reviews",
123   "_id" : "YXYfNSgBQuaETKQjIkME",
124   "_score" : 1.0,
125   "_source" : {
126     "ReviewTitle" : "Good work place",
127     "CompleteReview" : "Learnings and work life balance is
128     great .It is imperative that what sort of organization
129     in Honeywell you choose no universal salary structure
130     in India not with Honeywell globe .Learnings and work
131     life balance is great",
132     "Rating" : "4",
133     "Date" : "19-01-2018"
134   }
135 }
```

Create proper mapping index, new_reviews

Console Search Profiler Grok Debugger Painless Lab BETA

History Settings Help

200 - OK 699 ms

```
27- }
28-
29- GET /reviews/_mapping
30-
31- GET /reviews/_search
32- {
33-   "query": {
34-     | "match_all": {}
35-   }
36- }
37-
38- PUT /new_reviews
39- {
40-   "settings": {
41-     "number_of_shards": 2,
42-     "number_of_replicas": 1
43-   },
44-   "mappings": {
45-     "properties": {
46-       "CompleteReview": {
47-         "type": "text",
48-         "fields": {
49-           | "keyword": { "type": "keyword", "ignore_above": 256 }
50-         }
51-       },
52-       "Date": {
53-         "type": "date",
54-         "format": "dd-MM-yyyy"
55-       },
56-       "Rating": {
57-         "type": "integer"
58-       },
59-       "ReviewTitle": {
60-         "type": "text",
61-         "fields": { "keyword": { "type": "keyword" } }
62-       }
63-     }
64-   }
65- }
```

Click to send request

```
1- {
2-   "acknowledged" : true,
3-   "shards_acknowledged" : true,
4-   "index" : "new_reviews"
5- }
6-
```

Reindex data

Console Search Profiler Grok Debugger Painless Lab BETA

History Settings Help

200 - OK 241 ms

```
34- | "match_all": {}
35- }
36- }
37-
38- PUT /new_reviews
39- {
40-   "settings": {
41-     "number_of_shards": 2,
42-     "number_of_replicas": 1
43-   },
44-   "mappings": {
45-     "properties": {
46-       "CompleteReview": {
47-         "type": "text",
48-         "fields": {
49-           | "keyword": { "type": "keyword", "ignore_above": 256 }
50-         }
51-       },
52-       "Date": {
53-         "type": "date",
54-         "format": "dd-MM-yyyy"
55-       },
56-       "Rating": {
57-         "type": "integer"
58-       },
59-       "ReviewTitle": {
60-         "type": "text",
61-         "fields": { "keyword": { "type": "keyword" } }
62-       }
63-     }
64-   }
65- }
66-
67- POST _reindex?wait_for_completion=false
68- {
69-   "source": { "index": "reviews" },
70-   "dest": { "index": "new_reviews" }
71- }
```

Click to send request

```
1- {
2-   "task" : "HxIODGnVSXWSXSn9jsyOfA:31429"
3- }
4-
```

Track progress

ConsoleSearch ProfilerGrok DebuggerPainless LabBETA

HistorySettingsHelp

```
36- }
37-
38- PUT /new_reviews
39- {
40-   "settings": {
41-     "number_of_shards": 2,
42-     "number_of_replicas": 1
43-   },
44-   "mappings": {
45-     "properties": {
46-       "CompleteReview": {
47-         "type": "text",
48-         "fields": {
49-           "keyword": { "type": "keyword", "ignore_above": 256 }
50-         }
51-       },
52-       "Date": {
53-         "type": "date",
54-         "format": "dd-MM-yyyy"
55-       },
56-       "Rating": {
57-         "type": "integer"
58-       },
59-       "ReviewTitle": {
60-         "type": "text",
61-         "fields": { "keyword": { "type": "keyword" } }
62-       }
63-     }
64-   }
65- }
66-
67- POST _reindex?wait_for_completion=false
68- {
69-   "source": { "index": "reviews" },
70-   "dest": { "index": "new_reviews" }
71- }
72-
73- GET _tasks?action=*reindex&detailed
```

200 - OK225 ms

```
1- {
2-   "nodes": { }
3- }
4-
```

Click to send request

Delete old index to save space

ConsoleSearch ProfilerGrok DebuggerPainless LabBETA

HistorySettingsHelp

```
38- PUT /new_reviews
39- {
40-   "settings": {
41-     "number_of_shards": 2,
42-     "number_of_replicas": 1
43-   },
44-   "mappings": {
45-     "properties": {
46-       "CompleteReview": {
47-         "type": "text",
48-         "fields": {
49-           "keyword": { "type": "keyword", "ignore_above": 256 }
50-         }
51-       },
52-       "Date": {
53-         "type": "date",
54-         "format": "dd-MM-yyyy"
55-       },
56-       "Rating": {
57-         "type": "integer"
58-       },
59-       "ReviewTitle": {
60-         "type": "text",
61-         "fields": { "keyword": { "type": "keyword" } }
62-       }
63-     }
64-   }
65- }
66-
67- POST _reindex?wait_for_completion=false
68- {
69-   "source": { "index": "reviews" },
70-   "dest": { "index": "new_reviews" }
71- }
72-
73- GET _tasks?action=*reindex&detailed
74-
75- DELETE /reviews
```

200 - OK518 ms

```
1- {
2-   "acknowledged" : true
3- }
4-
```

Click to send request

Analizers:

Test default standard analyzer

Console Search Profiler Grok Debugger Painless Lab BETA

History Settings Help

200 - OK 77 ms

```
45- {
46-   "properties": {
47-     "CompleteReview": {
48-       "type": "text",
49-       "fields": {
50-         "keyword": { "type": "keyword", "ignore_above": 256 }
51-       }
52-     },
53-     "Date": {
54-       "type": "date",
55-       "format": "dd-MM-yyyy"
56-     },
57-     "Rating": {
58-       "type": "integer"
59-     },
60-     "ReviewTitle": {
61-       "type": "text",
62-       "fields": { "keyword": { "type": "keyword" } }
63-     }
64-   }
65- }
66-
67- POST _reindex?wait_for_completion=false
68- {
69-   "source": { "index": "reviews" },
70-   "dest": { "index": "new_reviews" }
71- }
72-
73- GET _tasks?action=*reindex&detailed
74-
75- DELETE /reviews
76-
77- POST /_analyze
78- {
79-   "text": ["Welcome to the GREAT wooolrd of ... Datascientest with
80-   "analyzer": "standard"
81- }
82-
```

Click to send request

```
1- {
2-   "tokens" : [
3-     {
4-       "token" : "welcome",
5-       "start_offset" : 0,
6-       "end_offset" : 7,
7-       "type" : "<ALPHANUM>",
8-       "position" : 0
9-     },
10-    {
11-      "token" : "to",
12-      "start_offset" : 8,
13-      "end_offset" : 10,
14-      "type" : "<ALPHANUM>",
15-      "position" : 1
16-    },
17-    {
18-      "token" : "the",
19-      "start_offset" : 11,
20-      "end_offset" : 14,
21-      "type" : "<ALPHANUM>",
22-      "position" : 2
23-    },
24-    {
25-      "token" : "great",
26-      "start_offset" : 15,
27-      "end_offset" : 20,
28-      "type" : "<ALPHANUM>",
29-      "position" : 3
30-    },
31-    {
32-      "token" : "wooolrd",
33-      "start_offset" : 21,
34-      "end_offset" : 29,
35-      "type" : "<ALPHANUM>",
36-      "position" : 4
37-    },
38-    {
39-      "token" : "of",
```

Add built-in analyzer to new_reviews

Console Search Profiler Grok Debugger Painless Lab BETA

History Settings Help

200 - OK 73 ms

```
64- }
65-
66-
67- POST _reindex?wait_for_completion=false
68- {
69-   "source": { "index": "reviews" },
70-   "dest": { "index": "new_reviews" }
71- }
72-
73- GET _tasks?action=*reindex&detailed
74-
75- DELETE /reviews
76-
77- POST /_analyze
78- {
79-   "text": ["Welcome to the GREAT wooolrd of ... Datascientest with
80-   "analyzer": "standard"
81- }
82-
83-
84- POST /new_reviews/_close
85- PUT /new_reviews/_settings
86- {
87-   "settings": {
88-     "analysis": {
89-       "analyzer": {
90-         "std_english": {
91-           "type": "simple",
92-           "stopwords": "_english_"
93-         }
94-       }
95-     }
96-   }
97- }
98- POST /new_reviews/_open
99- #check configured analyzer
100- GET /new_reviews/_settings
101-
```

Click to send request

```
1 # POST /new_reviews/_close
2 {
3   "acknowledged" : true,
4   "shards_acknowledged" : true,
5   "indices" : {
6     "new_reviews" : {
7       "closed" : true
8     }
9   }
10 }
11
12 # PUT /new_reviews/_settings
13 {
14   "acknowledged" : true
15 }
16
17 # POST /new_reviews/_open
18 {
19   "acknowledged" : true,
20   "shards_acknowledged" : true
21 }
22
23 # GET /new_reviews/_settings
24 {
25   "new_reviews" : {
26     "settings" : {
27       "index" : {
28         "routing" : {
29           "allocation" : {
30             "include" : {
31               "_tier_preference" : "data_content"
32             }
33           },
34         },
35         "number_of_shards" : "2",
36         "provided_name" : "new_reviews",
37         "creation_date" : "1753272768145",
38         "analysis" : {
39           "analyzer" : {
```

Experiment inline with char filters and token filters

Console Search Profiler Grok Debugger Painless Lab BETA

History Settings Help

200 - OK 509 ms

```
79 "text": ["Welcome to the GREAT wooworld of ... Datascience with
80   "Datascientest !!! :)"],
81 "analyzer": "standard"
82 }
83
84 POST /new_reviews/_close
85 PUT /new_reviews/_settings
86 {
87   "settings": {
88     "analysis": {
89       "analyzer": {
90         "std_english": {
91           "type": "simple",
92           "stopwords": "_english_"
93         }
94       }
95     }
96   }
97 }
98 POST /new_reviews/_open
99 #check configured analyzer
100 GET /new_reviews/_settings
101
102 GET /_analyze
103 {
104   "char_filter": [
105     {
106       "type": "mapping",
107       "mappings": [ "数据科学 => datascience" ]
108     },
109     "html_strip"
110   ],
111   "tokenizer": "standard",
112   "filter": [ "lowercase", "asciifolding" ],
113   "text": "I'm à <h1>great</h1> teacher - and I <strong>love
114   </strong> apprendre 数据科学!"
115 }
```

Click to send request

```
1- {
2-   "tokens": [
3-     {
4-       "token": "i'm",
5-       "start_offset": 0,
6-       "end_offset": 3,
7-       "type": "<ALPHANUM>",
8-       "position": 0
9-     },
10-    {
11-      "token": "a",
12-      "start_offset": 4,
13-      "end_offset": 5,
14-      "type": "<ALPHANUM>",
15-      "position": 1
16-    },
17-    {
18-      "token": "great",
19-      "start_offset": 10,
20-      "end_offset": 15,
21-      "type": "<ALPHANUM>",
22-      "position": 2
23-    },
24-    {
25-      "token": "teacher",
26-      "start_offset": 21,
27-      "end_offset": 28,
28-      "type": "<ALPHANUM>",
29-      "position": 3
30-    },
31-    {
32-      "token": "and",
33-      "start_offset": 31,
34-      "end_offset": 34,
35-      "type": "<ALPHANUM>",
36-      "position": 4
37-    },
38-    {
39-      "token": "i",
```

Build custom analyzer in its own index

Console Search Profiler Grok Debugger Painless Lab BETA

History Settings Help

200 - OK 405 ms

```
107   "mappings": [ "数据科学 => datascience" ]
108 },
109 "html_strip"
110 ],
111 "tokenizer": "standard",
112 "filter": [ "lowercase", "asciifolding" ],
113 "text": "I'm à <h1>great</h1> teacher - and I <strong>love
114   </strong> apprendre 数据科学!"
115 }
116
117 PUT /custom_analyzer
118 {
119   "settings": {
120     "analysis": {
121       "filter": {
122         "custom_stop": {
123           "type": "stop",
124           "stopwords": [ "apprendre" ],
125           "ignore_case": true
126         }
127       },
128       "char_filter": {
129         "chinese_mapping": {
130           "type": "mapping",
131           "mappings": [ "数据科学 => datascience" ]
132         }
133       },
134       "my_custom_analyzer": {
135         "type": "custom",
136         "char_filter": [ "html_strip", "chinese_mapping" ],
137         "tokenizer": "standard",
138         "filter": [ "lowercase", "custom_stop", "asciifolding" ]
139       }
140     }
141   }
142 }
143 }
```

Click to send request

```
1- {
2-   "acknowledged": true,
3-   "shards_acknowledged": true,
4-   "index": "custom_analyzer"
5- }
```


Test the custom analyzer

Console Search Profiler Grok Debugger Painless Lab BETA

History Settings Help

200 - OK 263 ms

```
113- "text": "I'm à <h1>great</h1> teacher - and I <strong>love
114-   </strong> apprendre 数据科学!"
115- }
116- PUT /custom_analyzer
117- {
118-   "settings": {
119-     "analysis": {
120-       "filter": {
121-         "custom_stop": {
122-           "type": "stop",
123-           "stopwords": [ "apprendre" ],
124-           "ignore_case": true
125-         },
126-       },
127-       "char_filter": {
128-         "chinese_mapping": {
129-           "type": "mapping",
130-           "mappings": [ "数据科学 => datascience" ]
131-         },
132-       },
133-       "analyzer": {
134-         "my_custom_analyzer": {
135-           "type": "custom",
136-           "char_filter": [ "html_strip", "chinese_mapping" ],
137-           "tokenizer": "standard",
138-           "filter": [ "lowercase", "custom_stop", "asciifolding" ]
139-         },
140-       },
141-     },
142-   },
143- }
144-
145- GET /custom_analyzer/_analyze
146- {
147-   "analyzer": "my_custom_analyzer",
148-   "text": "I'm à <h1>great</h1> teacher - and I <strong>love
149-     </strong> apprendre 数据科学!"
150- }
```

Click to send request

```
23- },
24- {
25-   "token": "teacher",
26-   "start_offset": 21,
27-   "end_offset": 28,
28-   "type": "<ALPHANUM>",
29-   "position": 3
30- },
31- {
32-   "token": "and",
33-   "start_offset": 31,
34-   "end_offset": 34,
35-   "type": "<ALPHANUM>",
36-   "position": 4
37- },
38- {
39-   "token": "i",
40-   "start_offset": 35,
41-   "end_offset": 36,
42-   "type": "<ALPHANUM>",
43-   "position": 5
44- },
45- {
46-   "token": "love",
47-   "start_offset": 45,
48-   "end_offset": 58,
49-   "type": "<ALPHANUM>",
50-   "position": 6
51- },
52- {
53-   "token": "datascience",
54-   "start_offset": 69,
55-   "end_offset": 73,
56-   "type": "<ALPHANUM>",
57-   "position": 8
58- }
59- }
60- }
61- }
```